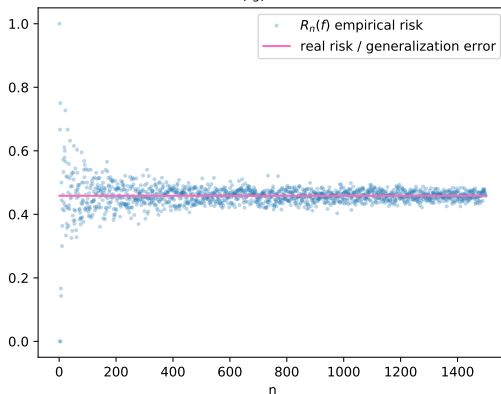


Fondamentaux théoriques du machine learning

f_3 : Empirical risk and generalization error
 $R(f_3)=0.46$



Exercise 1: (Analogous to the penalty shootout example) Consider the following random variable (X, Y) .

► $X \sim B(\frac{1}{2}),$

$$Y = \begin{cases} B(p) & \text{if } X = 1 \\ B(q) & \text{if } X = 0 \end{cases}$$

With $B(p)$ a Bernoulli law with parameter p .

► Hence $\mathcal{X} = \{0, 1\}, \mathcal{Y} = \{0, 1\}.$

Exercise 1: We consider 3 predictors :

- ▶ $f_1 : \{0, 1\} \rightarrow \{0, 1\}$:

$$f_1 = \begin{cases} 1 & \text{if } x = 1 \\ 0 & \text{if } x = 0 \end{cases}$$

- ▶ $\forall x, f_2(x) = 1 - f_1(x)$
- ▶ $\forall x, f_3(x) = 1$ We use the "0 - 1" loss.

Exercise 1 :

We observe the following dataset :

$$D_4 = \{(0, 1), (0, 0), (0, 0), (1, 0)\} \quad (1)$$

Compute the empirical risks $R_4(f_1)$, $R_4(f_2)$, $R_4(f_3)$.

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n l(y_i, f(x_i))$$

$$D_4 = \{(0, 1), (0, 0), (0, 0), (1, 0)\} \quad (2)$$

$$\begin{aligned} R_4(f_1) &= \frac{1}{4} \sum_{i=1}^4 l(f_1(x_i), y_i) \\ &= \frac{1}{4} \left(l(f_1(0), 1) + l(f_1(0), 0) + l(f_1(0), 0) + l(f_1(1), 0) \right) \\ &= \frac{1}{4} \times 2 \\ &= \frac{1}{2} \end{aligned} \quad (3)$$

$$D_4 = \{(0, 1), (0, 0), (0, 0), (1, 0)\} \quad (4)$$

$$\begin{aligned} R_4(f_2) &= \frac{1}{4} \sum_{i=1}^4 l(f_2(x_i), y_i) \\ &= \frac{1}{4} \left(l(f_2(0), 1) + l(f_2(0), 0) + l(f_2(0), 0) + l(f_2(1), 0) \right) \\ &= \frac{1}{4} \times 2 \\ &= \frac{1}{2} \end{aligned} \quad (5)$$

$$D_4 = \{(0, 1), (0, 0), (0, 0), (1, 0)\} \quad (6)$$

$$\begin{aligned} R_4(f_3) &= \frac{1}{4} \sum_{i=1}^4 l(f_3(x_i), y_i) \\ &= \frac{1}{4} \left(l(f_3(0), 1) + l(f_3(0), 0) + l(f_3(0), 0) + l(f_3(1), 0) \right) \\ &= \frac{1}{4} \times 3 \\ &= \frac{3}{4} \end{aligned} \quad (7)$$

Exercise 2: Consider the following random variable (X, Y) .

► $X \sim B(\frac{1}{2}),$

$$Y = \begin{cases} B(p) & \text{if } X = 1 \\ B(q) & \text{if } X = 0 \end{cases}$$

With $B(p)$ a Bernoulli law with parameter p .

► A predictor $f_1 : \{0, 1\} \rightarrow \{0, 1\} :$

$$f_1 = \begin{cases} 1 & \text{if } x = 1 \\ 0 & \text{if } x = 0 \end{cases}$$

With the "0 - 1" loss, what is the risk (generalization error) of f_1 , $R(f_1)$?

Exercise 2: Consider the following random variable (X, Y) .

► $X \sim B(\frac{1}{2}),$

$$Y = \begin{cases} B(p) & \text{if } X = 1 \\ B(q) & \text{if } X = 0 \end{cases}$$

► $f_1 : \{0, 1\} \rightarrow \{0, 1\} :$

$$f = \begin{cases} 1 & \text{if } x = 1 \\ 0 & \text{if } x = 0 \end{cases}$$

$$\begin{aligned} R(f_1) &= E[I(Y, f(X))] \\ &= 1 \times P(Y \neq f(X)) + 0 \times P(Y = f(X)) \\ &= P(Y \neq f(X)) \end{aligned} \tag{8}$$

► $X \sim B(\frac{1}{2}),$

$$Y = \begin{cases} B(p) & \text{if } X = 1 \\ B(q) & \text{if } X = 0 \end{cases}$$

► $f_1 : \{0, 1\} \rightarrow \{0, 1\} :$

$$f = \begin{cases} 1 & \text{if } x = 1 \\ 0 & \text{if } x = 0 \end{cases}$$

$$\begin{aligned} R(f_1) &= E[I(Y, f(X))] \\ &= 1 \times P(Y \neq f(X)) + 0 \times P(Y = f(X)) \\ &= P(Y \neq f(X)) \\ &= P((Y \neq f(X)) \cap (X = 1)) + P((Y \neq f(X)) \cap (X = 0)) \end{aligned} \quad (9)$$

$$\begin{aligned}R(f_1) &= E[l(Y, f(X))] \\&= 1 \times P(Y \neq f(X)) + 0 \times P(Y = f(X)) \\&= P(Y \neq f(X)) \\&= P((Y \neq f(X)) \cap (X = 1)) + P((Y \neq f(X)) \cap (X = 0)) \\&= P((Y \neq f(X)) | X = 1)P(X = 1) \\&\quad + P((Y \neq f(X)) | X = 0)P(X = 0)\end{aligned}\tag{10}$$

$$\begin{aligned} R(f_1) &= E[I(Y, f(X))] \\ &= 1 \times P(Y \neq f(X)) + 0 \times P(Y = f(X)) \\ &= P(Y \neq f(X)) \\ &= P((Y \neq f(X)) \cap (X = 1)) + P((Y \neq f(X)) \cap (X = 0)) \\ &= P((Y \neq f(X)) | X = 1)P(X = 1) \\ &\quad + P((Y \neq f(X)) | X = 0)P(X = 0) \\ &= \frac{1}{2}P((Y \neq 1) | X = 1) + \frac{1}{2}P((Y \neq 0) | X = 0) \end{aligned} \tag{11}$$

$$\begin{aligned}R(f_1) &= E[I(Y, f(X))] \\&= 1 \times P(Y \neq f(X)) + 0 \times P(Y = f(X)) \\&= P(Y \neq f(X)) \\&= P((Y \neq f(X)) \cap (X = 1)) + P((Y \neq f(X)) \cap (X = 0)) \\&= P((Y \neq f(X))|X = 1)P(X = 1) \\&\quad + P((Y \neq f(X))|X = 0)P(X = 0) \\&= \frac{1}{2}P((Y = 0)|X = 1) + \frac{1}{2}P((Y = 1)|X = 0) \\&= \frac{1}{2}(1 - p) + \frac{1}{2}q\end{aligned}$$

(12)

Exercise 3: Now consider

$$f_2 = \begin{cases} 0 & \text{if } x = 1 \\ 1 & \text{if } x = 0 \end{cases}$$

What is $R(f_2)$?

Exercice 3 :

$$\forall x, f_2(x) = 1 - f_1(x) \quad (13)$$

Exercise 3 :

$$\forall x, f_2(x) = 1 - f_1(x) \quad (14)$$

Hence

$$\begin{aligned} R(f_2) &= P(Y \neq f_2(X)) \\ &= P(Y \neq (1 - f_1(X))) \\ &= P(Y = f_1(X)) \\ &= 1 - R(f_1) \end{aligned} \quad (15)$$

Exercice 4 :

What is $R(f_3)$?

Exercice 4 :

$$\begin{aligned} R(f_3) &= P(Y \neq f_3(X)) \\ &= P(Y = 0) \end{aligned} \tag{16}$$

Exercise 4 :

$$\begin{aligned} R(f_3) &= P(Y \neq f_3(X)) \\ &= P(Y = 0) \\ &= P(Y = 0 \cap X = 0) + P(Y = 0 \cap X = 1) \\ &= P(Y = 0|X = 0)P(X = 0) + P(Y = 0|X = 1)P(X = 1) \\ &= \frac{1}{2}(1 - p) + \frac{1}{2}(1 - q) \end{aligned} \tag{17}$$